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Validation of bone surface modification models for inferring fossil hominin and carnivore feeding interactions, with reapplication to FLK 22, Olduvai Gorge, Tanzania☆

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ABSTRACT

Resolving the issue of how Early Stone Age hominins acquired large mammal carcasses requires information on their feeding interactions with large carnivores. This ecological information and its behavioral and evolutionary implications are revealed most directly from the tooth, cut, and percussion marks on bone surfaces generated by hominin and carnivore feeding activities. This paper employs a bootstrap method, a form of random resampling with replacement, to refine published neotaphonomic models that use the assemblage-wide proportions of long bones bearing feeding traces to infer the sequences in which Plio-Pleistocene hominins and carnivores accessed flesh, marrow, and/or grease from carcasses. Results validate the sensitivity of the models for inferring hominin feeding ecology, which have been questioned on grounds shown here to be unfounded. The bootstrapped feeding trace models are applied to the late Pliocene larger mammal fossil assemblage from FLK 22 (*Zinjanthropus* site), Olduvai Gorge, Tanzania. High frequencies of tooth and percussion marking on long bone midshaft fragments from FLK 22 are most consistent with those feeding trace models that simulate hominin scavenging from carcasses defleshed by carnivores, while cut mark data indicate that hominins more often had access to upper forelimb flesh than upper hind limb flesh. Together, the bone surface modification data indicate that hominins typically gained secondary access to partially defleshed carnivore kills, but they also allow for the possibility of some carcasses being processed only by carnivores and only by hominins.

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Introduction

Evidence of the timing of hominin and carnivore access to carcasses at Stone Age archaeological sites has important implications for understanding the adaptive capabilities of our ancestors during their initial and increasingly pervasive encroachment on the larger carnivore guild (e.g., Shipman and Walker, 1989; Blumenschine and Pobiner, 2007). Tooth, cut, and percussion marks on bone surfaces are the most direct and definitive traces for inferring the carnivorous component of fossil hominin diets and are also the clearest means for assessing ecological interactions with carnivores. As such, bone surface modifications are often the basis for interpretations of Stone Age archaeological bone assemblages, with the well preserved and large Oldowan assemblage from the FLK 22 (*Zinjanthropus* site; Leakey, 1971) receiving the majority of the attention (Binford, 1981, 1986, 1988; Bunn and Kroll, 1986, 1988; Oliver, 1994; Selvaggio, 1994, 1998; Blumenschine, 1995; Capaldo, 1995, 1997; Domínguez-Rodrigo, 1997; Marean and Kim, 1998; Marean et al., 2000; Egeland et al., 2004; Domínguez-Rodrigo and Barba, 2006; Blumenschine et al., 2007a).

Bunn (1981; Bunn and Kroll, 1986, 1988) was the first to use stone tool cut mark evidence for defleshing to suggest hominins had early access to carcasses at FLK 22. Binford (1986, 1988) countered Bunn by concluding that the FLK 22 hominins were only marginal scavengers of carcasses defleshed and ravaged by carnivores, basing his argument largely on the apparent dominance of lower over upper limb elements indicated by Leakey's (1971) skeletal part profiles from the assemblage. However, Bunn's interpretation lacked a statistical and actualistic foundation, and the skeletal part profiles used by Binford excluded midshaft fragments in long bone element counts (Bunn and Kroll, 1986). Subsequently, Blumenschine (1988, 1995) used the proportion of carnivore tooth-

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marked and hammerstone-on-anvil percussion-marked bovid long bone fragments derived from experimental butchery and marrow extraction, and observations of free-ranging carnivore carcass consumption in the Serengeti and Ngorongoro ecosystems to suggest that hominins at FLK 22 were scavenging from largely defleshed, abandoned carnivore kills.

The results of Blumenschine's statistical modeling of experimental bone assemblages defleshed and broken by stone tools and/or carnivores were later replicated and expanded upon by Marean (1991) with captive spotted hyenas; in the Serengeti ecosystem by both Selvaggio (1994) with a carnivore-followed-by-hominin scenario and Capaldo (1995) using the full range of skeletal parts; and by Lupo and O'Connell (2002) in an ethnographic context. Despite the large size (hundreds to thousands of specimens) of most of these samples, many of the models derived from them incorrectly use parametric statistics to describe nonparametric data. The models are nonparametric because each comprises a relatively small number of assemblages (groups of bones from a single individual and feeding episode/experimental trial), among which proportions of various feeding traces are not normally distributed. Marean et al. (2000) were the first to acknowledge the nonparametric distributions of the samples and to address the statistical shortcomings of the models using a bootstrap protocol. Their analysis was limited to Marean's sample and did not include Blumenschine's and Capaldo's data.

Efron (1979) introduced the bootstrap method and demonstrated the effectiveness of the technique on a variety of estimation problems. Tibshirani (1988:433) later noted that the bootstrap "provides a useful method for constructing confidence intervals for functional parameters in settings where likelihood methods are not applicable, particularly in nonparametric problems." The bootstrap is applied here to address the nonparametric distributions of assemblage means within the feeding trace models.

Some analyses of FLK 22 long bones have generated discrepant surface mark frequencies, and interpretation of the timing of hominin and carnivore access to carcasses is apparently counter-indicated by cut marks on the one hand, and tooth and percussion marks on the other. Specifically, cut mark data have been used to suggest hominin hunting or "power scavenging" from carnivore kills (e.g., Bunn and Kroll, 1986; Bunn and Ezzo, 1993; Dominguez-Rodrigo, 1997), while percussion and tooth mark data have been used to suggest secondary hominin access to partially defleshed carcasses (e.g., Blumenschine, 1995; Capaldo, 1997; Selvaggio, 1998; Blumenschine et al., 2007a). Clearly, there is a need to consider all three mark types simultaneously.

The original feeding trace models refined in this study have been used by a number of researchers to interpret feeding sequences at FLK 22 and other Paleolithic assemblages. The results obtained by these various studies and the sensitivity of the models to consumer sequence are now in question given the statistical limitations noted above. The value of the original feeding trace models and the bootstrap models presented here lies in their ability to distinguish between various sequences of hominin and carnivore carcass consumption. The models are controls grounded by actualistic observations and are intended to be used as comparative samples of known origin from which unknown feeding sequences associated with archaeological site formation can be inferred. The application here is to the FLK 22 assemblage, but the models are not specific to the site, nor should they be viewed as a simulation of it. As such, the greater significance of the models is that they are broadly applicable, allowing temporal and spatial comparisons between fossil assemblages.

We address the above mentioned issues by: 1) validating and refining the statistical models produced by Blumenschine (1995) and Capaldo (1995) using a bootstrap protocol; 2) incorporating a previously unexplored consumer sequence; 3) considering the carcass consumption sequence (Blumenschine, 1986a,b) when interpreting cut mark frequencies; and 4) applying the refined models to all surface traces of hominin and carnivore feeding found on long bones from FLK 22.

Methods

The statistics used here describe five distinct feeding trace models (Tables 1 and 2), four of which are published and named: Blumenschine's (1988, 1995) hammerstone-only (HO), carnivore-only (CO), and hammerstone-to-carnivore (H-C) models, and Capaldo's (1995) additional H-C sample, and whole-bone-to-carnivore (WB-C) model. The hammerstone-only and hammerstone-to-carnivore models are more appropriately labeled hominin-only and hominin-to-carnivore, respectively, because they also involve complete metal-knife defleshing (see below). A fifth scenario, vulture-to-hominin-to-carnivore (V-H-C), models a three and in some cases four-stage sequence of carcass consumption. This sample was previously included in Blumenschine's H-C sample but comprises a separate sample here due to the inclusion of cut mark frequencies not considered in his previous work. All models are based on only the major long bones, including the femur, humerus, tibia, radius-ulna, and metapodials. Long bones are divided into three portions: epiphyseal, near-epiphyseal, and midshaft portions, following Blumenschine (1988). None of the models describe hominin consumption of grease in cancellous long bone ends using boiling technology. As such, any model describing hammerstone breakage of long bones to extract marrow leaves both greasy bone ends still attractive to carnivores, and midshaft fragments that are devoid of edible tissues and ignored by scavengers (Blumenschine, 1988; Blumenschine and Marean, 1993; Capaldo, 1995; 1998).

One consumer type

The HO model describes hominins as the only consumer of a carcass, where all bone specimens were defleshed with a metal knife and broken using a hammerstone-on-anvil technique to remove marrow, producing assemblages of specimens with cut and percussion marks, but no tooth marks (details in Blumenschine, 1988). In the CO model, all bones were defleshed by various mammalian carnivores and then fragmented by spotted hyenas, and in one case lions, as they consumed marrow and grease from long bones, producing assemblages of specimens

Table 1
Nomenclature for feeding trace models used in this study[a]

Feeding Trace Models with Abbreviations	Foods Extracted from Long Bones		
	Flesh	Marrow	Grease
One Consumer Type			
Carnivore-Only (CO)	Carnivore	Carnivore	Carnivore
Hammerstone-Only (HO)	Hominin	Hominin	Not Applicable
Two Consumer Types (Hominin first)			
Hammerstone-to-Carnivore (H-C)	Hominin	Hominin	Carnivore
Whole-Bone-to-Carnivore (WB-C)	Hominin	Carnivore	Carnivore
Three Consumer Types			
Vulture-to-Hominin-to-Carnivore (V-H-C)	Vulture	Hominin	Carnivore

[a] Carnivores refer to larger mammalian carnivores only and in the Serengeti samples included lion and spotted hyena most commonly, and also cheetah and black-backed and golden jackals. Vultures (Ruppell's griffon, white-backed, hooded, lappet-faced) and raptors (tawny, Egyptian, and bateleur eagles) were frequently involved in the defleshing stage of the carnivore-only model. An alternative, two-consumer-type (carnivore first) model can be found in Selvaggio (1998), but the data as presented cannot be applied systematically here.

Table 2
Number of assemblages and specimens for each experimental model used in this study

Type of Feeding Trace Model	Number of Assemblages	Number of Specimens	
		Midshaft Fragments	All Long Bone Fragments
Carnivore-Only			
Size group 1–2	2	56	64
Size group 3–4	7	118	167
Size group 1–4 (Total)	9	174	231
Hammerstone-Only			
Size group 1–2	6	212	303
Size group 3–4	1	15	24
Size group 1–4 (Total)	7	227	327
Whole-Bone-to-Carnivore			
Size group 1–2	7	60	72
Size group 3–4	4	95	121
Size group 1–4 (Total)	11	155	193
Hammerstone-to-Carnivore			
Size group 1–2	32	1,079	1,350
Size group 3–4	14	597	799
Size group 1–4 (Total)	46	1,676	2,149
Vulture-to-Hominin-to-Carnivore			
Size group 1–2	1	17	23
Size group 3–4	4	93	124
Size group 1–4 (Total)	5	110	147
Grand Total	78	2,342	3,047

with tooth marking as the only feeding trace (details in Blumenschine, 1988).

Two consumer types

The H-C model simulates carnivore scavenging of carcasses from which hominins had consumed all flesh and marrow, producing assemblages with all three types of feeding traces. Unlike Blumenschine's long bone sample ($n = 7$ assemblages), Capaldo's also included crania, ribs, vertebrae, pelves, and scapulae ($n = 39$ assemblages). Only long bone fragments are included in this analysis, and we combine Blumenschine's and Capaldo's samples because their published mean tooth and percussion mark frequencies differ by less than three percentage points.

In Capaldo's WB-C model, bones were defleshed with a metal knife before carnivores fragmented them to access marrow and grease, producing assemblages with cut and tooth marks. Capaldo conducted 30 WB-C experiments, but only 19 were included in his analysis due to the complete consumption or removal of bone by carnivores in the others. The sample discussed here is further reduced to 11 experiments by excluding those with sample sizes of less than five bone fragments in order to limit the affect of small sample sizes on mean assemblage values.

Three consumer types

The V-H-C sample is composed of five assemblages using long bones from four carcasses, all of which were fed upon by vultures. Some may have been defleshed minimally by carnivores, and one had been defleshed substantially by carnivores prior to their discovery. Most exhibited moderately to highly fat-depleted marrow (Sinclair and Duncan, 1972; Blumenschine and Madrigal, 1993), resulting in reduced or no spotted hyena consumption of hammerstone-generated epiphyseal fragments. Minimal carnivore defleshing is evidenced by at most marginal gnawing on processes of only proximal long bones. Subsequently, bones were disarticulated with a metal knife and hammerstone-broken to extract all marrow before being left on the landscape for carnivores. As such, specimens in this model collectively bear carnivore tooth marks, in some cases from defleshing and in others from breakage of hammerstone-generated epiphyseal fragments. The specimens in this model can also bear cut marks on epiphyseal fragments and percussion marks on all long bone portions. Despite the involvement of vultures in the defleshing of carcasses, beak and/or claw marks were not observed on any of the V-H-C specimens. Further, raptor damage is usually characterized by "can opener" perforations (Sanders et al., 2003:99) or digestive rounding and thinning of bone (Robert and Vigne, 2002), with neither being commonly found on larger mammal long bones or easily mistaken for the diagnostic pits and scores, v-shaped straie, and microstriations that precisely characterize tooth, cut, and percussion marks, respectively (Blumenschine et al., 1996). While these assemblages are grouped into one sample due to the small number of constituent assemblages, they are variable in carcass size, type of initial consumers, and the number and type of elements included, as follows:

Assemblage 1 Assemblage 1 consists of a sub-adult, animal size group 3 wildebeest (*Connochaetes taurinus*) with moderately fat-depleted marrow that was discovered with virtually no flesh on the bones and with the metapodials covered with skin. Scavenging opportunities included the brain and all other within-bone nutrients. It was likely killed and fed upon by lions (inferred on the basis of marginal gnawing on some long bone epiphyses) prior to removal of remaining flesh by vultures. Blumenschine disarticulated all long bones and extracted all marrow before placing the hammerstone-broken fragments on the landscape for scavengers.

Assemblage 2 This assemblage consists of a sub-adult, animal size group 4 buffalo (*Syncerus caffer*) that was discovered by Blumenschine after all flesh had been consumed. Only the tibia is included in the sample. The buffalo had been highly nutritionally stressed as seen in its low-fat marrow. The absence of gnawing on even the femoral greater trochanter and the tuberosities of the ischia and humeri indicates that vultures had likely consumed all flesh. The tibia was hammerstone-broken to remove all marrow, and resulting bone fragments were placed on the landscape for carnivores to exploit.

Assemblages 3 and 4 These assemblages contain an adult, animal size group 3 wildebeest with extremely fat-depleted marrow. Upon discovery, ten vultures were feeding and a single spotted hyena was in the vicinity. All flesh had been consumed and only the lesser tuberosity of one humerus exhibited marginal carnivore gnawing typically coincident with flesh removal by felids, in this case most likely lions. All limbs were collected and hammerstone-broken. Two bone fragment clusters were placed in different landscape settings for carnivores to scavenge. Assemblage 3 comprised the right forelimb fragments, while assemblage 4 comprised those of the left forelimb and left hind limb. In both cases, tooth marking was minimal and epiphyses were left unconsumed by scavengers, an apparent result of the poor condition of the animal at death.

Assemblage 5 Assemblage 5 consists of a juvenile Thompson's gazelle (*Gazella thomsonii*) that is the only carcass representing animal size group 1 in the V-H-C sample. Three jackals, the likely predators, and 20 vultures were observed to feed on the carcass. Unlike other V-H-C assemblages, this is the only one to have been defleshed largely by mammalian carnivores. Subsequent defleshing by vultures justifies the inclusion of the assemblage in the V-H-C sample. Vultures are capable of completely consuming small flesh scraps left by carnivores that would otherwise have been available for hominins to scavenge, thereby depressing cut mark frequencies. Remaining food comprised all podial skin, flesh scraps on the humerus and radius, and within-bone nutrients. After hammerstone breakage, all bones were left for carnivore scavengers to consume.

The FLK 22 assemblage

Data presented in this paper for the incidences of tooth-, percussion-, and cut-marked bones from FLK 22 are based on Blumenschine's (1995) analysis of the assemblage. Details on the coding of marks and the sample (NISP = 731) used can be found in Blumenschine (1995) for tooth and percussion marks, and in Capaldo (1997) for cut marks. The tooth and percussion mark frequencies for FLK 22 presented here were taken from Blumenschine's (1995) Table 3, while the cut mark data were generated anew from his database. Data for the incidence of specimens both tooth-marked and cut- and/or percussion-marked (tooth- and butchery-marked throughout) were also generated from Blumenschine's database. This class of marked bone was used by Selvaggio (1994, 1998) to describe specimens bearing feeding traces produced by both hominins and carnivores.

Bootstrap methods

A bootstrap sample must reflect the structure of the original sample in order to estimate the sample distribution (Tibshirani, 1988; Efron and Tibshirani, 1993). In the case of this study, there are five models, comprising *N* assemblages (groups of bones from a single individual and feeding episode/trial) that comprise *n* specimens. Thus, the bootstrap algorithm used here accounts for the number of assemblages within each model and the number of specimens within each assemblage.

The bootstrap algorithm

A bootstrap algorithm developed using R version 2.8.1 (The R Foundation for Statistical Computing, 2008) was applied to the data on the presence and absence of bone surface modifications (tooth, percussion, and cut marks) within the modern control assemblages. This algorithm consisted of three functions that generated bootstrap distributions of assemblage bone modification proportions for each model. The first function is designed to provide equal potential weight to each bone within a given assemblage. The second function provides equal potential weight to each assemblage within a given model and weights the bootstrap samples according to the number of assemblages within each feeding trace model. The third function replicates the first two 10,000 times. The three functions used are explained in detail below using the WB-C model as an example.

The first function selects an assemblage at random from a given feeding trace model. For example, one assemblage would be selected at random from the eleven that comprise the WB-C model. The data for the assemblage are selected and are equal to bone modification type, in this case the presence or absence of tooth marks on each specimen. The function then determines the number of specimens (*n*) that are within the selected assemblage and resamples the data (tooth marks) with replacement *n* times to create a bootstrap replicate of the assemblage. A specimen can be selected more than once or not at all, providing equal potential weight to each specimen. The final step of this function is to obtain the mean proportion of specimens that were modified, in this case tooth-marked. At this stage, obtaining the mean in the algorithm serves to model the proportion of specimens from a single assemblage that are tooth-marked.

The second function applies the first function *N* times, with *N* being equal to the number of assemblages in the model that is being resampled. For the WB-C model, the first function is applied eleven times, and any of the eleven assemblages that comprise the WB-C model may be selected more than once or not at all, providing equal potential weight to each assemblage. This generates eleven different means for the proportion of tooth-marked bone that are then averaged into a single mean. At this stage there is a single bootstrap replicate of the WB-C model. The reduction of variability that results from taking the mean of means is necessary for the bootstrap algorithm to accommodate the disparity in the number of assemblages among models. Had the second function not accounted for the number of assemblages in a given model, variability would have increased with sample size, rather than decrease as expected.

The third function applies the second function 10,000 times, producing 10,000 randomly generated means, which are then used to calculate the summary statistics on central tendency and dispersion. At this stage, the WB-C model has been replicated 10,000 times. This process is repeated for each model and type of bone modification (tooth marks, cut marks, percussion marks, and tooth and butchery marks). In the case of the WB-C model, this included tooth, cut, and tooth and butchery marks, but not percussion marks as the model describes bone breakage by carnivores rather than hominins.

Samples subdivided by animal size groups and long bone portion represent separate samples of 10,000 means, rather than a portion of the original bootstrap sample. The bootstrap samples for each bone modification type were generated independently, such that the tooth- and butchery-marked class was created as a separate category within the original non-randomized database prior to applying the bootstrap.

Interpreting bootstrap distributions

The bootstrap distributions we present model 10,000 possible outcomes that represent the behavioral processes that created the original non-randomized feeding trace models. As such, had a second bootstrap been conducted, the 10,000 values generated and the summary statistics describing their distribution would differ slightly from the first. It is also important to note that the bootstrap distributions only reflect the conditions (e.g., habitat, number/type/hunger of carnivore consumers, and butchery tools used) under which the original experiments were generated.

Standard box and whisker plots and 95% interquantile ranges are used to depict the bootstrap distributions. The 95% interquantile ranges are based on the 2.5% and 97.5% percentiles for the distributions of means in the bootstrap samples and can more generally be referred to as bootstrap confidence intervals (Efron and Tibshirani, 1993). The 95% interquantile ranges are used to assess differences among models at the 0.05 level of probability and can be interpreted much like a traditional 95% confidence interval. Here, when the 95% interquantile ranges overlap among models, the models are considered to be indistinguishable. When they do not overlap they are considered as likely distinct.

Results

Percussion marks

The incidences of percussion marking from the HO, H-C, and V-H-C models are statistically indistinguishable from one another with one exception: the 95% interquantile ranges for all size group 3–4 long bone fragments in the HO and H-C models do not overlap (Table 3; Fig. 1a–c). Despite overlap among the size group sub-samples and among the bone portion sub-samples of the bootstrapped models, the mean incidence of percussion marking in the HO model is generally slightly higher than in the H-C and V-H-C models. The only exception is for size group 1–2 midshaft fragments, where the H-C model has the highest mean incidence of percussion-marked specimens.

Table 3
Incidence of percussion-marked bone generated by bootstrap analysis[a]

	All Long Bone Fragments				Midshaft Fragments			
	HO	H-C	V-H-C	FLK 22	HO	H-C	V-H-C	FLK 22
Size group 1–4								
Mean %	38.5	26.5	24.9	27.4	27.7	24.9	19.8	27.1
Median ± I.Q.R.	38.5 ± 6.6	26.5 ± 3.5	24.3 ± 12.1	–	27.9 ± 7.5	24.9 ± 3.6	19.5 ± 10.3	–
Min.–Max.	20.4–55.9	16.8–36.3	1.5–62.3	–	3.3–46.6	14.3–36.1	0–51.9	–
95% Interquantile Range	29.1–47.7	21.6–31.6	9.1–43.6	–	16.4–37.9	19.8–30.2	6.4–35.8	–
Size group 1–2								
Mean %	36.6	30.3	17.5	32.4	26.7	29.4	23.6	32.4
Median ± I.Q.R.	36.6 ± 7.0	30.2 ± 4.3	17.4 ± 8.7	–	26.9 ± 8.4	29.4 ± 4.3	23.5 ± 11.8	–
Min.–Max.	18.3–54.6	17.9–43.3	0–56.5	–	0–45.6	19.2–42.1	0–70.6	–
95% Interquantile Range	26.3–46.2	24.3–36.6	4.4–34.8	–	14.0–37.8	23.6–35.8	5.9–47.1	–
Size group 3–4								
Mean %	49.9	18.0	26.9	25.3	33.5	14.6	19.2	24.8
Median ± I.Q.R.	50.0 ± 16.7	17.8 ± 4.7	26.4 ± 15.2	–	33.3 ± 13.3	14.3 ± 5.0	18.7 ± 12.4	–
Min.–Max.	8.3–83.3	7.4–33.8	1.9–69.2	–	0–80.0	3.8–29.9	0–58.0	–
95% Interquantile Range	29.2–70.8	11.5–25.5	7.7–49.1	–	13.3–60.0	8.1–22.4	3.7–38.3	–

[a] Incidence is measured as the proportion of specimens bearing at least one percussion mark. HO, Hammerstone-Only; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore; I.Q.R., Interquartile range; 95% Interquantile range, values between the 2.5% and 97.5% quantiles. Data for the FLK 22 assemblage is taken from Table 3 of Blumenschine (1995). Highlighted boxes indicate where FLK 22 is within the 95% interquantile ranges and therefore statistically indistinguishable from the model.

The incidence of percussion marking in the FLK 22 assemblage falls within the 95% interquantile ranges of all size group and bone portion sub-samples of the bootstrapped V-H-C model. FLK 22 is slightly below the 95% interquantile ranges of all long bone fragments from both the combined size groups and size group 3–4 of the HO model, and is slightly above them for size group 3–4 midshaft fragments of the H-C model.

Tooth marks

The incidence of tooth marking differentiates the CO and WB-C models from the H-C and V-H-C models for all long bone fragments and midshaft fragments of the combined size groups (Table 4; Fig. 2a). However, when broken into size group sub-samples, the discriminatory power of the models is weakened by overlap in the relatively large 95% interquantile ranges for all long bone fragments and midshaft fragments of the size group 1–2 sub-samples of the V-H-C model (Fig. 2b), and for midshaft fragments of the size group 3–4 sub-sample of the WB-C model (Fig. 2c).

For all size groups combined and for size group 3–4, the incidence of tooth marking in the FLK 22 assemblage is outside the 95% interquantile ranges for nearly all sub-samples of all models. The exception is for midshaft fragments of all size groups combined and of size group 3–4, where FLK 22 is within the 95% interquantile ranges of the WB-C model. For all long bone fragments and midshaft fragments of size group 1–2, the value for the FLK 22 assemblage falls within the 95% interquantile ranges of the CO and WB-C models, and is marginally above the 95% interquantile range of the V-H-C model.

Cut marks

The incidence of cut marking does not differentiate most sub-samples of the bootstrapped models (Table 5; Fig. 3a–c). For all size group 3–4 long bone fragments, the 95% interquantile ranges of the H-C and V-H-C models overlap but are statistically distinct from the intervals of the overlapping HO and WB-C models. For the V-H-C model, cut marking is absent in the size group 1–2 sub-sample of all long bone fragments, and in all size group sub-samples of midshaft fragments.

The incidence of cut marking in the FLK 22 assemblage falls within the 95% interquantile ranges of all sub-samples of only the H-C model. For all long bone fragments, the value for FLK 22 also falls within the 95% interquantile ranges of the WB-C model for all size groups combined and for the size group 1–2 sub-sample. For all size group sub-samples of midshaft fragments, the value for FLK 22 is within the 95% interquantile ranges of both the HO and WB-C models, but not the V-H-C model.

Tooth and butchery marks

The incidence of specimens bearing both tooth and butchery marks does not differentiate the bootstrapped models for most sub-samples (Table 6; Fig. 4a–c). The exception is for all long bone fragments of the size group 3–4 sub-sample, where the 95% interquantile range of the WB-C model is above that of the H-C model. Generally, the WB-C model has the highest mean incidence of specimens bearing both tooth and butchery marks. The only exception is for midshaft fragments of the size group 1–2 sub-sample, where the V-H-C model has the highest value.

The incidence of specimens bearing both tooth and butchery marks in the FLK 22 assemblage is higher than the 95% interquantile ranges of most sub-samples of the WB-C and V-H-C models, and all sub-samples of the H-C model. For all long bone fragments, the value for FLK 22 falls within the 95% interquantile ranges of the combined size groups and the size group 3–4 sub-sample of the WB-C model. For midshaft fragments, the value for FLK 22 falls within the 95% interquantile ranges of the size group 1–2 sub-sample of the V-H-C model and the size group 3–4 sub-sample of the WB-C model.

Discussion

The bootstrap method refines and validates the feeding trace models

The results of random resampling with replacement of data from feeding trace models validate Blumenschine's (1988, 1995) and Capaldo's (1995) published claims of the models' ability to discriminate the general identity, sequence, and tissues extracted by consumers of larger mammal long bone flesh, marrow, and grease. The CO model is distinguished from all others in having very high tooth mark frequencies along with an absence of percussion and cut marks. The HO model is distinguished from all others in having high percussion and cut mark frequencies, but an absence of

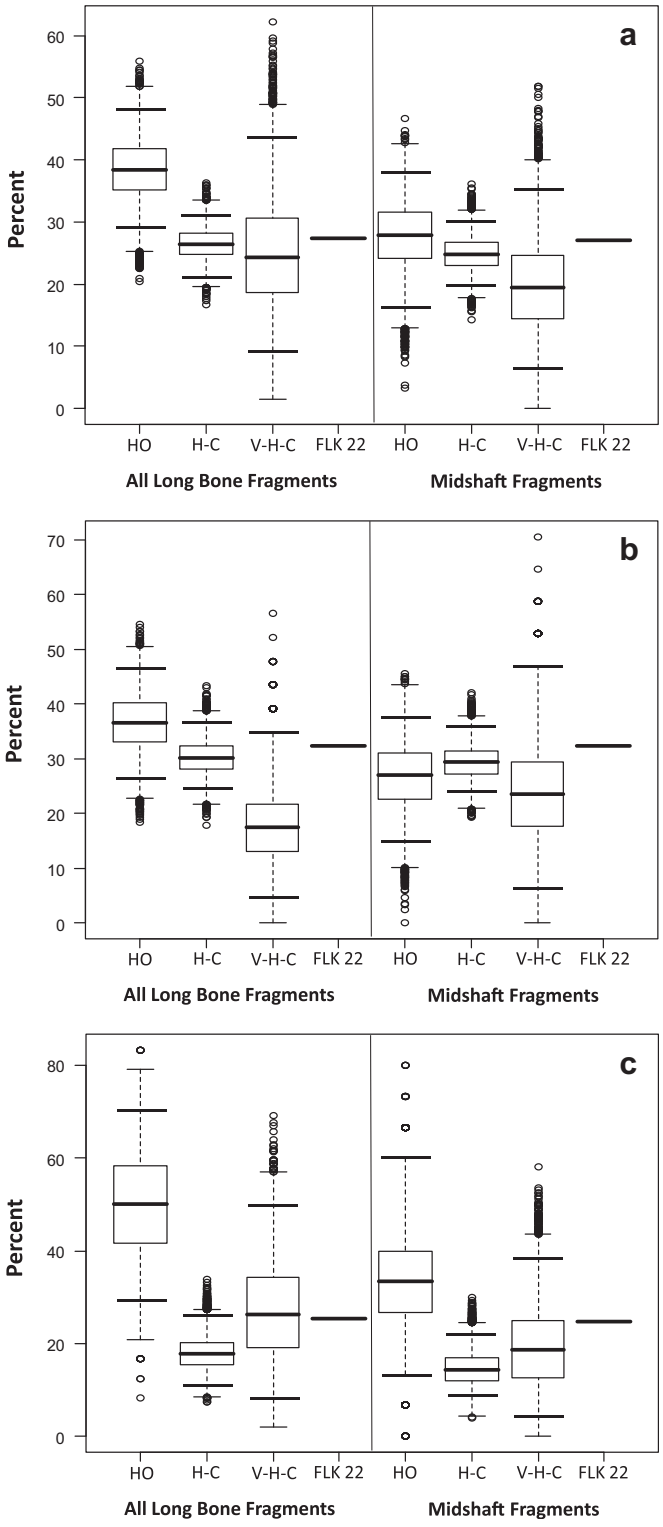


Fig. 1. Incidence of percussion-marked bone for (a) animal size group 1–4, (b) animal size group 1–2, and (c) animal size group 3–4. HO, Hammerstone-Only; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore. For the experimental feeding trace models the bold center lines represent the median; the lower boxes are the second quartiles; the upper boxes are the third quartiles; the bold longer lines beyond the boxes represent the 95% interquantile ranges; and the whiskers represent either the maximum value or 1.5 times the interquartile range. Any data outside 1.5 times the interquartile ranges are outliers and are depicted as circles. For FLK 22 the bold center line represents the proportion of percussion-marked bone.

carnivore tooth marks. The 95% interquantile ranges for both of these models closely approximate the 95% confidence intervals reported by Blumenschine (1995). The H-C model differs from the previous two in bearing all three types of marks and is distinct from the WB-C model in the latter's absence of percussion marks. Further, the H-C model is distinct from the V-H-C model because it has cut marks on midshaft fragments not present in the latter. Blumenschine's and Capaldo's published 95% confidence intervals for H-C are similar to the 95% interquantile range that describes their combined H-C samples. Capaldo's published 95% confidence intervals for the WB-C model are slightly smaller than the 95% interquantile ranges, which describe a sample reduced by the exclusion of assemblages with fewer than five specimens.

The bootstrap also improves the discriminatory power of subsamples of the models, the usefulness of which were limited previously by large 95% confidence intervals resulting from small sample sizes. For example, the bootstrap reduced the interval indicative of significance at the 0.05 level within the size group 1–2 sub-sample of the CO model from 33.1 to 100% tooth-marked to 56.7–82.5% tooth-marked. Additionally, the bootstrap approach allows the use of 95% interquantile ranges to describe models comprised of a single assemblage (i.e., size group 3–4 of the HO model and size group 1–2 of the V-H-C model). This last point applies with the caveat that bootstrap distributions built from a single assemblage may reflect fewer conditions affecting rates of bone modification than those built from multiple assemblages.

Finally, the bootstrap has refined the models by allowing us to detect a subdivision of the H-C model on the basis of whether vultures or hominins removed the flesh. The V-H-C model was originally lumped together with the H-C model by Blumenschine (1995) because he was not considering cut marks, having used a metal knife instead of stone flakes to deflesh the bones. Modeled cut mark frequencies may differ in unknown ways if a stone flake had been used instead of a metal knife for butchery. Yet, the low or complete absence of cut marking for all size group and bone portion sub-samples of the V-H-C model differentiates it from the H-C model and also from the HO and WB-C models. In particular, cut marks within the V-H-C model result from disarticulation only, as reflected by their absence on midshaft fragments. Additionally, some assemblages in the V-H-C model were observed or inferred from marginal bone gnawing to have experienced some defleshing by carnivores prior to or coincident with vulture access. In one case, assemblage 5 (the only size group 1–2 assemblage), jackals were observed to have consumed all but flesh scraps from the long bones. This produced the relatively high mean incidence of tooth marking on long bone midshafts in assemblage 5 when compared with size group 1–2 of the H-C model and size group 3–4 of the V-H-C model (Fig. 2). In fact, the incidence of tooth marking in Selvaggio's (1998) carnivore-to-hominin-to-carnivore model, of which assemblage 5 is an example, lies within the upper end of the 95% interquantile range for this assemblage but is beyond the range of all other models' size group 1–2 sub-samples.

Unwarranted doubts surrounding the use of feeding trace models

The validity and discriminatory power of feeding trace models have been questioned by Lupo and O'Connell (2002) and Faith (2007). For example, Faith (2007:1602) states,

> "Doubts surrounding the usefulness of bone surface modifications stem from a large and currently inexplicable range of variation across experimental and archaeological assemblages. It is clear that unless the sources of variability can be identified and corrected for in our analytical frameworks, bone surface damage patterns will continue to be problematic as indicators of

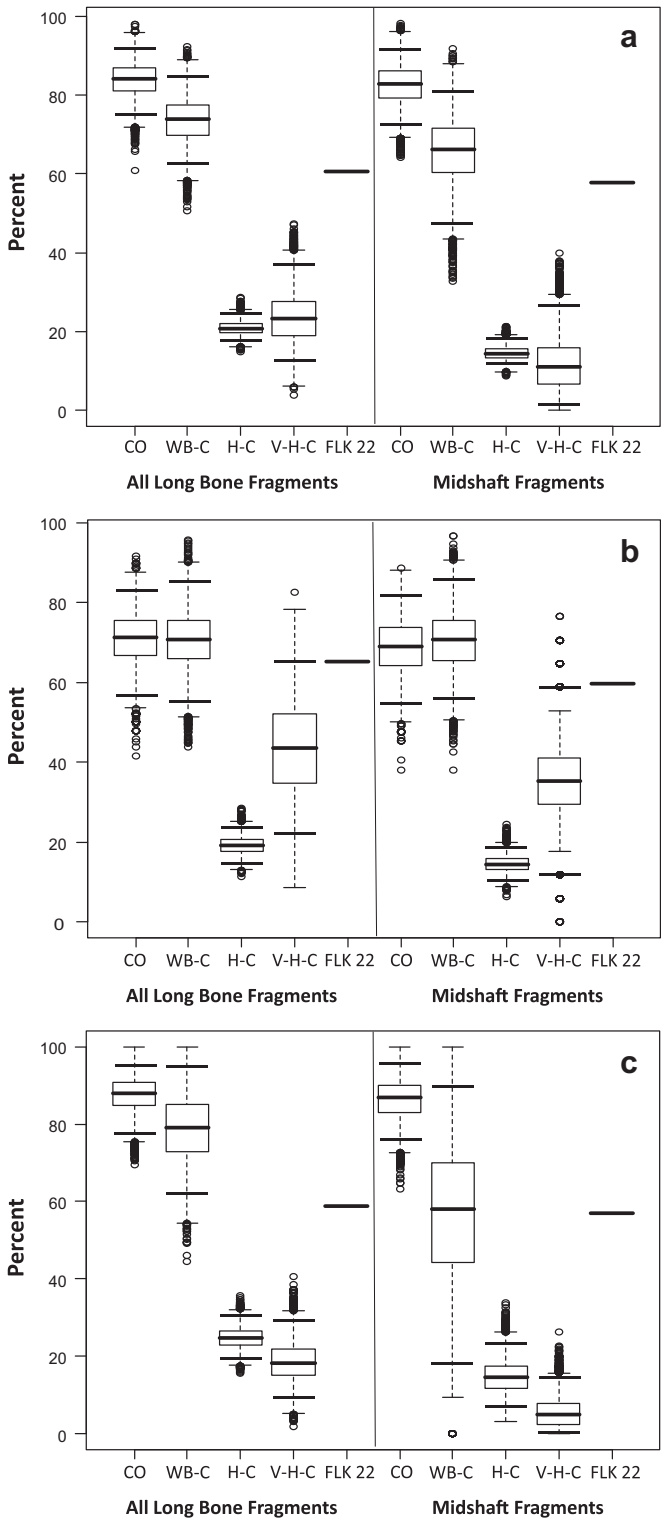


Fig. 2. Incidence of tooth-marked bone for (a) animal size group 1–4, (b) animal size group 1–2, and (c) animal size group 3–4. CO, Carnivore-Only; WB-C, Whole-Bone-to-Carnivore; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore. For the experimental feeding trace models the bold center lines represent the median; the lower boxes are the second quartiles; the upper boxes are the third quartiles; the bold longer lines beyond the boxes represent the 95% interquantile ranges; and the whiskers represent either the maximum value or 1.5 times the interquartile range. Any data outside 1.5 times the interquartile ranges are outliers and are depicted as circles. For FLK 22 the bold center line represents the proportion of tooth-marked bone.

the timing and impact of human and carnivore agents on archaeological bone assemblages."

Yet the source of variability is known, reported unambiguously in the description of the flesh consumers and bone breakers in the original published definition of each model (Table 1). These descriptions show that Lupo and O'Connell's and Faith's characterizations of the models as "human-first," "carnivore-first," or "carnivore only" are oversimplified, thereby masking the full range of behavioral processes (defleshing, bone breakage, and/or grease removal) simulated by the various models. The descriptions also reveal that Lupo and O'Connell (2002: Fig. 3), Faith (2007: Fig. 1), and others (Marean et al., 2000: Table 3; Domínguez-Rodrigo and Barba, 2006: 171) have misclassified the WB-C model variously as "carnivore first" or "carnivore only." As well, they lump into "carnivore first" both Selvaggio's (1994, 1998) carnivore (defleshing only) to hominin (defleshing and marrow removal) model and Blumenschine's CO model.

The WB-C is a "human first" not "carnivore first" or "carnivore only" model because it describes total flesh removal with a metal knife prior to carnivore access (see Capaldo, 1995: 55–58; 1997: 565–566; 1998: 314–315). The absence of defleshing by carnivores depresses tooth mark frequencies below those in the CO model, but they remain significantly higher than the H-C model because the WB-C model describes carnivore breakage of epiphyseal ends and midshaft cylinders. Likewise, the lower rate of tooth marking on Selvaggio's (1998) carnivore-to-hominin model (65.7%) relative to that on CO results from an absence of bone-breaking tooth marks. Hence, cut mark values covary with the tooth mark values: the lower rate of tooth marking on both the WB-C model and Selvaggio's carnivore-to-hominin model relative to CO is matched by the presence of cut marks in the former pair, but not CO, and additionally for Selvaggio's model by the presence of percussion marks. Similarly, the higher tooth mark values on WB-C relative to H-C result from carnivore breakage of intact long bones in the former versus only hammerstone-generated epiphyseal fragments in the latter. Conversely, the WB-C model lacks percussion marks present in the H-C model.

When comparisons of mark frequencies are limited to independently produced models that describe the same actions of hominins and carnivores, a remarkably high degree of correspondence is evident. Hence, Marean's and Blumenschine's CO models produced overall mean tooth marking frequencies within six percentage points of one another, while their and Capaldo's H-C model have a three-way correspondence of mean tooth mark frequencies that differ by less than seven percentage points. Therefore, the claim of a large range of variation across comparable models is false.

Claims of low inter-analyst correspondence in surface mark estimates of the FLK 22 assemblage (Lupo and O'Connell, 2002; Domínguez-Rodrigo and Barba, 2006) are also unwarranted (Blumenschine, 1995; Blumenschine et al., 2007a). Lupo and O'Connell (2002) cite statistical differences between tooth and cut mark frequencies on long bone specimens recorded independently by Blumenschine (1995) and Oliver (1994). Oliver reported a lower frequency of tooth-marked bone than Blumenschine. He also reported a higher frequency of cut-marked bone than Blumenschine, whose data were first reported by Selvaggio (1994) and Capaldo (1997). Lupo and O'Connell (2002:97, 99) correctly hypothesized that either the researchers did not use the same samples in their analyses, or the criteria used for the identification of tooth and cut marks were defined differently. In fact, both the samples analyzed and the criteria used for the identification of bone surface modifications, described fully by Blumenschine (1995) and Oliver (1994) within the methods sections of their respective

Table 4
Incidence of tooth-marked bone generated by bootstrap analysis[a]

	All Long Bone Fragments					Midshaft Fragments				
	CO	WB-C	H-C	V-H-C	FLK 22	CO	WB-C	H-C	V-H-C	FLK 22
Size group 1–4										
Mean %	83.9	73.7	20.1	23.6	60.7	82.6	65.7	14.5	11.7	57.9
Median ± I.Q.R.	84.1 ± 6.0	73.9 ± 7.7	20.9 ± 2.4	23.3 ± 8.6	–	82.8 ± 6.8	66.2 ± 11.2	14.5 ± 2.3	11.0 ± 9.1	–
Min.–Max.	60.9–98.0	50.6–92.2	14.8–28.6	4.0–47.3	–	64.3–98.3	32.9–91.8	8.7–21.3	0–40.0	–
95% Interquantile Range	74.6–92.1	62.3–84.5	17.4–24.6	12.2–37.0	–	72.5–91.9	47.8–80.8	11.4–18.1	1.3–26.2	–
Size group 1–2										
Mean %	70.9	70.6	19.1	43.4	65.3	69.1	70.5	14.5	35.5	59.7
Median ± I.Q.R.	71.3 ± 8.8	70.8 ± 9.7	19.1 ± 3.0	43.5 ± 17.4	–	69.1 ± 9.5	70.7 ± 10.0	14.4 ± 2.8	35.3 ± 11.8	–
Min.–Max.	41.7–91.7	44.0–95.6	11.4–28.6	8.7–82.6	–	38.1–88.6	38.1–96.8	6.4–24.4	0–76.5	–
95% Interquantile Range	56.7–82.5	55.7–84.3	14.9–23.7	21.7–65.2	–	54.8–81.0	56.1–84.7	10.5–18.9	11.8–58.8	–
Size group 3–4										
Mean %	87.6	78.9	24.9	18.5	58.9	86.5	57.3	14.8	5.6	57.1
Median ± I.Q.R.	88.0 ± 6.2	79.1 ± 12.2	24.9 ± 3.6	18.3 ± 6.7	–	86.8 ± 7.0	57.9 ± 25.7	14.5 ± 5.8	5.0 ± 5.3	–
Min.–Max.	69.4–100	44.5–100	15.7–35.7	1.9–40.6	–	63.1–100	0–100	3.2–33.9	0–26.3	–
95% Interquantile Range	77.9–95.5	61.9–95.0	19.6–30.3	9.1–28.7	–	75.7–95.8	18.1–89.8	7.3–23.5	0–14.4	–

[a] Incidence is measured as the proportion of specimens bearing at least one tooth mark. CO, Carnivore-Only; WB-C; Whole-Bone-to-Carnivore; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore; I.Q.R., Interquartile range; 95% Interquantile range, values between the 2.5% and 97.5% quantiles. Data for the FLK 22 assemblage is taken from Table 3 of Blumenschine (1995). Highlighted boxes indicate where FLK 22 is within the 95% interquantile ranges and therefore statistically indistinguishable from the model.

papers, differ considerably. Oliver's (1994) analysis was based on a larger sample than Blumenschine's, probably because Blumenschine explicitly excluded specimens smaller than 2 cm in maximum length, as well as those with poor surface visibility and/or recent breaks. Oliver's inclusion of specimens smaller than 2 cm and with poor surface visibility would likely depress frequencies of all marks, while his inclusion of specimens with major recent breaks would have an unpredictable effect on tooth and cut mark frequencies. More importantly, Oliver (1994:272) was overly conservative in his identification of tooth marks stating,

> "pits and scores define carnivore activity only if (1) the two co-occur on the same area of bone, (2) one or the other occur in large numbers so that there is little doubt that potential mimics could not have produced the marks, and/or (3) they are associated with other more diagnostic carnivore-induced damages such as furrows, punctures and lever-up breaks."

These criteria will always produce underestimates of tooth mark frequencies compared to those such as Blumenschine's and Capaldo's that include less conspicuous and solitary tooth marks. Specimens with such marking are common in control collections. Blumenschine et al. (1996) have shown that they are morphologically diagnostic and reliably identified in blind testing by both experienced analysts and by novices with as little as 2-h experience with control collections.

More recently, Domínguez-Rodrigo and Barba (2006) argued that the incidence of tooth marking on long bone midshaft fragments from FLK 22 is significantly lower (11%) than the numbers reported by Blumenschine (1995: 57.9%). They claim that Blumenschine (1995) confused the assemblage's previously undescribed "biochemical marks" produced by microorganisms with carnivore tooth marks, resulting in a revised estimate of tooth mark proportions consistent with the H-C model. However, Blumenschine et al. (2007a) show that Domínguez-Rodrigo's and Barba's (2006) assertion that microbes will bioerode bone surfaces in ways that mimic carnivore tooth marks is unsubstantiated due to general methodological flaws that have yet to be remedied.

When Oliver's (1994) and Domínguez-Rodrigo's and Barba's (2006) analyses are excluded, strong inter-analyst correspondence of tooth mark estimates at FLK 22 is evident. In fact, Blumenschine's

Table 5
Incidence of cut-marked bone generated by bootstrap analysis[a]

	All Long Bone Fragments					Midshaft Fragments				
	HO	WB-C	H-C	V-H-C	FLK 22	HO	WB-C	H-C	V-H-C	FLK 22
Size group 1–4										
Mean %	29.9	27.0	18.3	6.7	18.7	11.7	25.0	14.4	0	15.9
Median ± I.Q.R.	29.8 ± 5.4	26.6 ± 10.2	18.3 ± 2.6	6.4 ± 5.3	–	11.6 ± 5.0	24.4 ± 12.6	14.3 ± 2.5	0 ± 0	–
Min.–Max.	15.6–46.3	6.6–57.8	12.0–27.5	0–23.4	–	0–24.8	1.6–68.8	8.6–23.0	0	–
95% Interquantile Range	22.2–38.3	13.7–42.8	14.8–22.3	0–15.3	–	4.5–19.3	9.1–44.6	11.0–18.3	0	–
Size group 1–2										
Mean %	27.2	18.4	18.9	0	17.8	10.3	16.4	15.4	0	14.4
Median ± I.Q.R.	27.3 ± 4.7	17.7 ± 10.3	18.8 ± 3.4	0 ± 0	–	10.3 ± 4.9	15.5 ± 11.9	15.2 ± 3.3	0 ± 0	–
Min.–Max.	13.6–39.0	0–53.4	11.4–31.8	0	–	0–23.9	0–58.1	6.9–28.6	0	–
95% Interquantile Range	20.3–33.7	5.5–34.8	14.4–24.4	0	–	3.1–17.6	2.5–36.0	10.9–20.6	0	–
Size group 3–4										
Mean %	45.9	41.9	16.8	8.5	19.1	20.1	40.0	12.2	0	16.5
Median ± I.Q.R.	45.8 ± 16.7	40.9 ± 18.4	16.7 ± 3.4	8.1 ± 6.3	–	20.0 ± 13.3	39.4 ± 29.5	12.2 ± 3.1	0 ± 0	–
Min.–Max.	12.5–83.3	11.4–95.0	7.9–29.5	0–29.7	–	0–66.7	0–100	4.1–21.2	0	–
95% Interquantile Range	25.0–66.7	22.1–69.5	12.1–21.9	0–18.4	–	0–40.0	13.0–80.1	7.9–16.9	0	–

[a] Incidence is measured as the proportion of specimens bearing at least one cut mark. HO, Hammerstone-Only; WB-C, Whole-Bone-to-Carnivore; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore; I.Q.R., Interquartile range; 95% Interquantile range, values between the 2.5% and 97.5% quantiles. Data for the FLK 22 assemblage is reworked from Blumenschine's (1995) analysis. Highlighted boxes indicate where FLK 22 is within the 95% interquantile ranges and therefore statistically indistinguishable from the model.

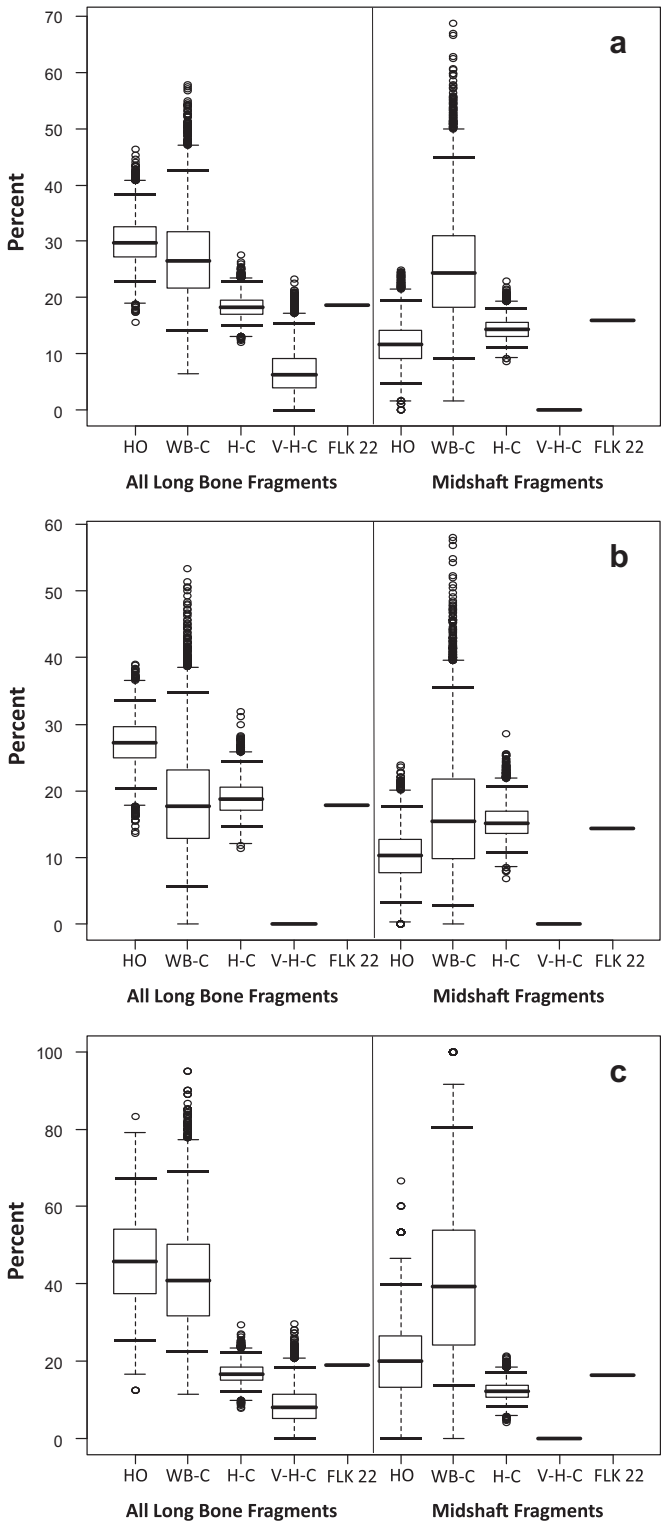


Fig. 3. Incidence of cut-marked bone for (a) animal size group 1–4, (b) animal size group 1–2, and (c) animal size group 3–4. HO, Hammerstone-Only; WB-C, Whole-Bone-to-Carnivore; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore. For the experimental feeding trace models the bold center lines represent the median; the lower boxes are the second quartiles; the upper boxes are the third quartiles; the bold longer lines beyond the boxes represent the 95% interquantile ranges; and the whiskers represent either the maximum value or 1.5 times the interquartile range. Any data outside 1.5 times the interquartile ranges are outliers and are depicted as circles. For FLK 22 the bold center line represents the proportion of cut-marked bone.

(1995) estimate for the frequency of tooth marking in the FLK 22 assemblage is the only one that has been independently replicated. Capaldo also recorded relatively high tooth mark frequencies on long bone midshaft fragments (49.0%; see Blumenschine et al., 2007b). The slightly lower incidence of tooth marking observed by Capaldo is most likely related to the larger sample that he used (n = 1,153). Many (28.0%) of the additional specimens Capaldo included are small bone fragments (2–3 cm in maximum length), which have been shown to be tooth-marked at lower frequencies than their larger counterparts (Blumenschine, 1988, 1995; Faith, 2007). Blumenschine's and Capaldo's similar estimates arise because both analysts have extensive experience with bone tooth-marked by carnivores under controlled conditions. Both also employ the same published criteria for identifying conspicuous and inconspicuous tooth marks, use the same standards to create the analytical sample of specimens from FLK 22 upon which mark frequencies are based, and have been successfully blind-tested on bone surface mark identification (Blumenschine et al., 1996). Most importantly, both independent tooth mark frequency estimates are intermediate to the 95% interquantile ranges of the H-C and CO models, indicating that neither model describes the dominant feeding interactions at FLK 22.

Implications for hominin and carnivore behavior at FLK 22

Results indicate the pattern of feeding traces found at FLK 22 cannot be accounted for by some of the models. Most clearly, the co-occurrence in the assemblage of tooth-marked specimens and butchery-marked specimens eliminates individually the CO and/or HO models as exclusive descriptors of the behavioral sequences that produced the FLK 22 assemblage. Likewise, individual specimens bearing both tooth and butchery marks eliminate a combination of CO and HO as exclusive descriptors.

Results also indicate the pattern of feeding traces found at FLK 22 can be accounted for by some of the behavioral sequences modeled by the experimental assemblages. Other sequences thus far unexplored by feeding trace modelers might also account for the FLK 22 pattern. For example, we cannot discount hominins as tooth mark producers on at least some bones (White and Toth, 2007). Hominin tooth marking might be expected principally on long bone fragments of size group 1–2 animals but is less likely to contribute to the frequency of tooth-marked bones from larger animals. In fact, size group 1–2 long bones from FLK 22 are tooth-marked more frequently than their size group 3–4 counterparts, suggesting the possibility that some of the tooth marks were inflicted by hominins. Still, the models available are ecologically realistic (Blumenschine, 1986b) and are consistent with what is known of Plio-Pleistocene East African carnivore guild structure (Werdelin and Lewis, 2005).

The bootstrap results support Blumenschine's (1995) contention that the incidence of percussion marking in the FLK 22 assemblage indicates hominins to have been the primary consumers of marrow at the site. FLK 22 is within or close to the 95% interquantile ranges for all of the site formation models that describe primary access to marrow by hominins. When compared with the HO model, the slightly lower incidence of percussion marking in the FLK 22 assemblage for all long bone fragments, and the nearly identical incidence of percussion marking for midshaft fragments, is indicative of bone-crunching carnivores having deleted from the site or partly consumed percussion-marked epiphyseal and near-epiphyseal specimens from bones that were hammerstone-broken by hominins. The incidence of percussion marking at FLK 22 is most similar to the H-C model for size group 1–2 and the V-H-C model for size group 3–4.

Table 6
Incidence of tooth- and butchery-marked bone generated by bootstrap analysis[a]

	All Long Bone Fragments				Midshaft Fragments			
	WB-C	H-C	V-H-C	FLK 22	WB-C	H-C	V-H-C	FLK 22
Size group 1–4								
Mean %	19.7	8.4	9.6	25.2	9.7	4.9	4.4	24.5
Median ± I.Q.R.	19.2 ± 8.0	8.3 ± 1.8	9.2 ± 6.3	–	9.4 ± 5.1	4.9 ± 1.5	4.0 ± 4.4	–
Min.–Max.	4.3–52.1	4.5–14.7	0–35.0	–	0–27.8	1.5–10.0	0–22.9	–
95% Interquantile Range	9.5–32.2	6.0–11.4	2.0–19.7	–	3.0–18.0	2.9–7.4	0–12.2	–
Size group 1–2								
Mean %	11.6	8.6	8.7	30.1	8.2	5.9	11.6	21.9
Median ± I.Q.R.	11.1 ± 6.5	8.5 ± 2.5	8.7 ± 8.7	–	7.4 ± 6.9	5.8 ± 2.2	11.8 ± 11.8	–
Min.–Max.	0–36.7	3.7–18.8	0–43.5	–	0–34.3	1.5–14.5	0–47.1	–
95% Interquantile Range	3.3–22.4	5.4–12.5	0–21.7	–	0–19.7	3.2–9.4	0–29.4	–
Sizes group 3–4								
Mean %	33.7	8.0	9.9	23.2	12.3	3.2	2.5	22.7
Median ± I.Q.R.	32.6 ± 14.9	7.9 ± 1.8	9.3 ± 7.4	–	12.0 ± 7.5	3.1 ± 1.5	2.5 ± 5.0	–
Min.–Max.	7.3–85.0	3.6–14.1	0–36.5	–	0–41.0	0.2–9.2	0–22.5	–
95% Interquantile Range	16.7–57.5	5.5–10.9	1.6–22.3	–	2.6–24.4	1.3–5.8	0–10.0	–

[a] Incidence is measured as the proportion of specimens bearing at least one tooth and one percussion and/or cut mark. WB-C, Whole-Bone-to-Carnivore; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore; I.Q.R., Interquartile range; 95% Interquantile range, values between the 2.5% and 97.5% quantiles. Data for the FLK 22 assemblage is reworked from Blumenschine's (1995) analysis. Highlighted boxes indicate where Zinj is within the 95% interquantile ranges and therefore statistically indistinguishable from the model.

The incidence of tooth marking in the FLK 22 assemblage is consistent with carnivores, not hominins, having had primary access to flesh. FLK 22 has a far greater incidence of tooth marking than is predicted by the H-C model. The similar incidences of tooth marking at FLK 22 and in the WB-C model, which simulates hominin defleshing of carcasses prior to carnivore scavenging, is not surprising, nor does it counter the hypothesis that carnivores were the primary consumers of flesh at the site. Defleshed but unbroken long bone shafts are highly attractive to bone-crunching carnivores, which in exposing marrow cavities produce tooth-marked fragments at frequencies only slightly lower than those described by the CO model. However, fragmentation and extraction of marrow by hominins (i.e., the H-C and V-H-C models) leaves nutritionally unattractive midshaft fragments; the relatively few tooth-marked midshaft fragments are produced through carnivore fragmentation of hammerstone-generated epiphyseal fragments when they subsequently access grease. Of course, the WB-C model as the dominant process is rejected because it lacks percussion marks, which are present in high frequencies in the FLK 22 assemblage.

The incidence of tooth marking on size group 1–2 midshaft fragments at FLK 22 is only slightly above the 95% interquantile range for the one assemblage (5) of like-sized specimens from the V-H-C model. Because this assemblage represents carnivore defleshing, this result is fully consistent with indications from tooth marks described above. The incidence of tooth marking on long bone midshaft fragments for all size groups from FLK 22 (57.9%) is also similar to the mean incidence of tooth marking for midshaft fragments from Selvaggio's (1998) carnivore-hominin model (47.0%) and especially her carnivore-to-hominin-to-carnivore model (54.2%), both of which also describe carnivores having primary access to flesh and hominins having primary access to marrow.

The relatively high incidence of cut marking in the FLK 22 assemblage is consistent with hominins having had access to flesh on many skeletal parts, but this does not contradict the tooth marking results. The frequency of cut-marked bone in the FLK 22 assemblage is nearly identical to that for all sub-samples of the H-C model, which simulates hominins having had sole access to flesh, and is higher than predicted by all sub-samples of the V-H-C model. Particularly interesting is the high incidence of cut marking on long bone midshafts from FLK 22 and the absence of cut-marked midshafts in the V-H-C model, indicating many of the cut marks in the FLK 22 assemblage may be related to defleshing. However, whether this reflects defleshing of whole muscle masses or only the still attractive scraps that typically remain after mammalian carnivore defleshing is uncertain: cut mark frequencies have not been shown to be sensitive to the amount of flesh remaining on bones (Capaldo, 1998; Pobiner and Braun, 2005; Blumenschine and Pobiner, 2007).

The apparent contradiction between tooth and cut marking is resolved by examining the distribution of cut marks among flesh-bearing long bone elements at FLK 22 (Table 7). A chi-square test of association demonstrates that the incidence of cut marking on humeral midshaft fragments (39.5%) is significantly greater than on femoral midshaft fragments (12.5%; $p = 0.02$, $\chi^2 = 5.19$, d.f. $= 1$). The same pattern was noted by Oliver (1994:280) for all long bone fragments, but his interpretation of the data was focused on the "preferential placement of cut marks on meaty limbs," speculating that the paucity of cut marks on the femur resulted from a deficiency of femoral ends in the assemblage. The above data show this difference is apparent despite the exclusion of long bone epiphyses. It is also predicted by Blumenschine's (1986a,b) carcass consumption sequence, which demonstrates that carnivores typically consume the higher yielding upper hindquarter flesh prior to forequarter flesh. As well, tibia midshaft fragments are cut-marked more frequently (33.3%) than midshafts from the radius/ulna (19.6%), but, as expected from their near equivalent ranking in the carcass consumption sequence, this difference is not significant at the 0.05 level. Overall, the pattern of cut marking in the FLK 22 assemblage is consistent with hominins typically having access to humerus flesh more often than femur flesh. This indicates that hominins were typically acquiring carcasses after carnivores had defleshed at least the upper hindquarters, and possibly the lumbar vertebrae and the rib cage. This would yield approximately 50% of total flesh weight of bovids if the remaining parts were fully fleshed (Blumenschine and Caro, 1986), or much lower amounts if they had been partially consumed. These interpretations apply with the caveat that differences in skeletal and soft tissue anatomy between the forelimb and hind limb may influence cut mark frequencies in currently unknown ways.

The high incidence of specimens bearing both tooth and butchery marks for FLK 22 supports hominins having typically acquired carcasses partially defleshed by felids. Such high incidences are only described by models that have high frequencies of tooth marks on long bone midshaft fragments, including the WB-C model, assemblage 5 of the V-H-C model, and Selvaggio's (1998)

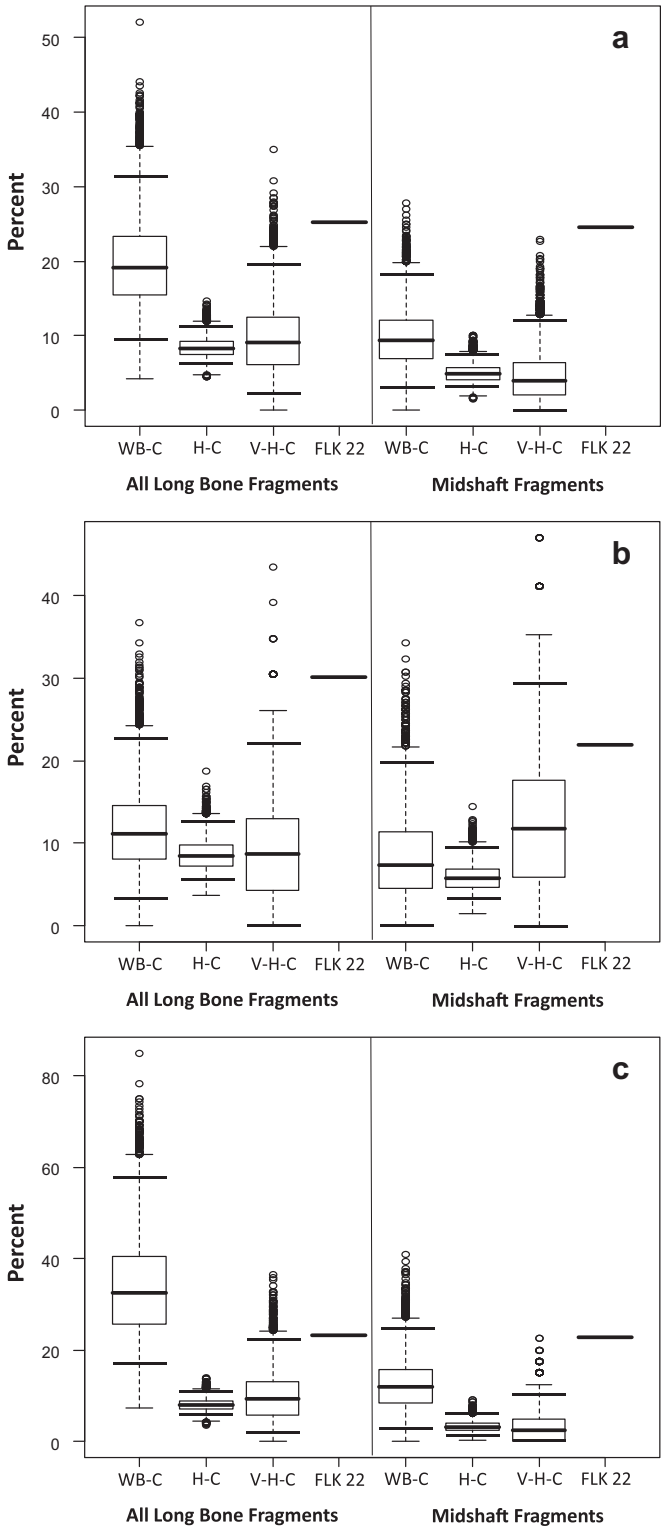


Fig. 4. Incidence of tooth- and butchery-marked bone for (a) animal size group 1–4, (b) animal size group 1–2, and (c) animal size group 3–4. WB-C, Whole-Bone-to-Carnivore; H-C, Hammerstone-to-Carnivore; V-H-C, Vulture-to-Hominin-to-Carnivore. For the experimental feeding trace models the bold center lines represent the median; the lower boxes are the second quartiles; the upper boxes are the third quartiles; the bold longer lines beyond the boxes represent the 95% interquantile ranges; and the whiskers represent either the maximum value or 1.5 times the interquartile range. Any data outside 1.5 times the interquartile ranges are outliers and are depicted as circles. For FLK 22 the bold center line represents the proportion of tooth- and butchery-marked bone.

Table 7
Incidence of cut-marked flesh-bearing midshaft fragments within the FLK 22 assemblage by long bone element[a]

Element	Cut-Marked	Total	% Cut-Marked
Femur	3	24	12.5
Humerus	15	38	39.5
Tibia	20	60	33.3
Radius–Ulna	11	56	19.6

[a] Incidence is measured as the proportion of specimens bearing at least one cut mark. Data generated from Blumenschine's (1995) analysis. Elements are listed in order of their ranking in the carcass consumption sequence from earliest consumed to latest (Blumenschine, 1986a,b).

carnivore-to-hominin and carnivore-to-hominin-to-carnivore models (which have 95% confidence intervals of 15.0–36.0% and 18.0–43.2%, respectively, for the incidence of specimens bearing tooth and butchery marks). However, all of the butchery marks for the WB-C model are cut marks, while those from FLK 22 also include percussion marks, such that only the remaining models that describe carnivore defleshing prior to hominin access fit the co-occurring tooth and butchery mark data.

The high incidence of specimens bearing both tooth and butchery marks at FLK 22 does not allow rejection of a common amenity scenario for the assemblage, as has been argued by Egeland et al. (2004). The high incidence of tooth marks suggests a possible CO assemblage component, while the high incidences of cut and percussion marks suggest a possible HO component. However, these components would involve the minority of carcasses, with most being best described by the carnivore-to-hominin-to-carnivore model. On the other hand, a high incidence of tooth- and butchery-marked specimens could result from hominins having had variable points of initial access in the carcass consumption sequence. Hominin access early in the consumption sequence of some carcasses could produce a dominance of butchery over tooth marks on these bones, while late access could produce dominantly carnivore tooth-marked bone. All of these scenarios are consistent with new information that specifies the paleolandscape setting of FLK 22, including a lightly wooded peninsula at which hominins might have acquired tree-stored leopard kills, and an adjacent wetland at which carnivores seem to have ambushed prey and left scavengeable carcasses that could be exploited by hominins (cf. Blumenschine et al., 2012).

Finally, a low incidence or absence of midshaft specimens bearing both tooth and butchery marks does not provide evidence that hominins and carnivores fed on different carcasses. For example, Egeland et al. (2004) claim that the low incidence of such specimens in the early Pleistocene large mammal assemblage from Swartkrans (2.3%) implies such independent feeding by hominins and carnivores. Instead, the ranges of the WB-C (0–27.8%), H-C (1.5–10%), and V-H-C (0–22.9%) models accommodate the Swartkrans value despite the dual consumption of all carcasses by hominins and carnivores they describe.

Conclusions

Refinement and expansion of feeding trace models produced by Blumenschine (1995) and Capaldo (1995) through resampling validates the primacy of bone surface modification studies for inferring the behavioral ecology of hominin carnivory. We have shown that published criticisms of these models are unfounded and that the models can be used to test hypotheses about hominin subsistence ecology. Other interpretive criteria such as skeletal part and age profiles (Klein, 1980; Shipman et al., 1981; Cruze-Uribe, 1991) have not been shown to be sensitive to the order in which hominins and carnivores accessed carcasses largely because they are influenced heavily by density-dependent destruction of bone

(Marean et al., 1992; Lam et al., 1998). As such, the frequency of bone surface modifications should be emphasized in behavioral analyses of archaeological assemblages bearing traces of both hominin and carnivore carcass consumption. This prescription comes with the caveat that analysts reporting surface mark frequencies use the same criteria for excluding specimens from which surface mark frequencies are to be calculated, have experience with collections of bones marked by known consumers under controlled conditions, use published criteria for differentiating marks, and have demonstrated the reliability of their identifications through blind testing using control collections.

Doubts surrounding the usefulness of bone surface modification models have stalled their further development by other researchers for nearly a decade. The results presented here justify their refinement with the goal of expanding the range of modeled behaviors, environments, and consumers sequences.

The integration of tooth, cut, and percussion mark data, and consideration of the carcass consumption sequence to explain cut mark distributions, provides evidence for a more specific scenario of hominin and carnivore carcass consumption for the FLK 22 assemblage than was possible previously. Percussion mark frequencies for FLK 22 suggest that hominins broke the majority of long bones. Tooth and cut mark data indicate both hominins and carnivores had access to flesh, although whether hominin access was to entire muscle masses or only scraps, as suggested by Selvaggio's (1994, 1998) models, is uncertain. Finally, the high incidence of specimens that are both tooth- and butchery-marked demonstrate frequent hominin and carnivore feeding from the same carcasses.

Together these data are consistent with a dominant three-stage carnivore-to-hominin-to-carnivore model of site formation. In the first feeding stage, carnivores (most likely felids) typically consumed at least upper hindquarter flesh and possibly all axial flesh posterior to the cervical vertebrae before abandoning the carcass or being displaced from it by hominins. In the second feeding stage, hominins typically acquired carcasses affording all within-bone tissues and at least widespread flesh scraps if not whole muscle masses usually from the forequarters, neck, and head. Whether this was through passive scavenging of abandoned kills (cf. Blumenschine, 1986b; Cavallo and Blumenschine, 1989), or through active appropriation of attended, but partially consumed, carcasses (cf. Bunn and Ezzo, 1993; O'Connell et al., 2003) cannot be ascertained from the feeding trace models currently available. In the third and final feeding stage, bone-crunching carnivores (most likely hyaenids) consumed grease within hammerstone-broken long bone epiphyseal fragments and marrow from any long bone shafts left unbroken by hominins. This three-stage feeding trace model explains the at-first contradictory tooth (Selvaggio, 1994; 1998; Blumenschine, 1995; Capaldo, 1995; 1997) and cut (Domínguez-Rodrigo, 1997) mark data, and is consistent with the lack of evidence for hunting technology in the Oldowan.

The evolutionary significance of conclusions based on the bootstrap results for FLK 22 lies in the indication that the initial encroachment by hominins on the larger carnivore guild during the Oldowan may have been in the role of a scavenger. However, application of the feeding trace models to additional sites of similar age is necessary to determine whether the results for the FLK 22 assemblage are specific to the site or are indicative of the subsistence capabilities of our Plio-Pleistocene ancestors. Conversely, the few published applications of these models to later time periods are best described by the H-C model including results for assemblages from JK2 and WK, Beds III and IV, Olduvai Gorge (Pante, 2010), the Middle Paleolithic site of Kobeh, Iran (Marean and Kim, 1998), and the Middle Stone Age South African Cave sites Die Kelders Cave 1 (Marean et al., 2000), Blombos Cave, and Pinnacle Point Cave 13B (Thompson, 2008), suggesting a more dominant hominin role in the carnivore guild characterized by either early confrontational scavenging or hunting. Thus, the models, if more broadly applied, will allow paleoanthropologists to track the increasingly pervasive role hominins played in the larger carnivore guild throughout the Plio-Pleistocene, an evolutionary transition that has culminated in modern humans' super-predator status in most terrestrial and some aquatic ecosystems.

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